ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** force x displacement	W = Fs.
2	**(c)** changes the direction of force	MA = 1.
3	**(c)** 20 J	Fs.
4	**(c)** joule	J.
5	**(c)** pivot of a lever	Fixed point.
6	**(b)** 20 W	W/t.
7	**(c)** watt	W.
8	**(a)** first class lever	Fulcrum in the middle.
9	**(b)** 100 J	mgh.
10	**(b)** work/time	Rate.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force multiplied by displacement in the direction of the force (2).	2
12	Energy is neither created nor destroyed (2).	2
13	Rigid bar that turns about a fixed point (2).	2
14	Rate of doing work (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moving car (1); Flowing water (1); Flying bird (1).	1+1+1
16	W = Fs (1); 20 x 5 (1); 100 J (1).	1+1+1
17	PE to KE (1); Total constant (1); Without air resistance (1).	1+1+1
	OR: Stretched bow (1); Water in a tank (1); Book on a shelf (1).	1+1+1
18	First: scissors (1); Second: wheelbarrow (1); Third: tweezers (1).	1+1+1
	OR: Motion vs position (1); mv squared vs mgh (1); Both in joules (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Force (1); Displacement (1); In the direction of force (1); Examples (1); Zero work cases (1).	1+1+1+1+1
	OR: Ramp (1); Longer path (1); Less force (1); MA = l/h (1); Examples (1).	1+1+1+1+1
20	W = Fs (1); F = ma (1); v squared = 2as (1); W = (1/2) m v squared (1); Equal to KE (1).	1+1+1+1+1
	OR: Statement (1); Pendulum (1); KE and PE (1); Total constant (1); Friction (1).	1+1+1+1+1

Mistakes to avoid:

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** second class lever	Load in the middle.
2	**(b)** 16 J	0.5 x 2 x 16.
3	**(d)** (1/2) m v squared	Formula.
4	**(c)** third class lever	Effort in the middle.
5	**(a)** zero	No displacement.
6	**(a)** m g h	Formula.
7	**(b)** inclined plane	Machine.
8	**(d)** 1 kWh	0.1 x 10.
9	**(d)** potential energy	Position.
10	**(b)** easier	Convenience.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Work done when a force of 1 N moves a body 1 m (2).	2
12	Load divided by effort (2).	2
13	Energy due to motion (2).	2
14	Power when 1 J of work is done in 1 s (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Electric bulb (1); Fan (1); Solar cell (1).	1+1+1
16	Lift heavy loads (1); Change direction of force (1); Reduce effort (1).	1+1+1
17	KE to PE and back (1); Total constant (1); Air resistance reduces it (1).	1+1+1
	OR: 0.5 x 4 x 25 (1); = 50 (1); J (1).	1+1+1
18	No displacement (1); W = Fs (1); s = 0 (1).	1+1+1
	OR: Heat (1); Light (1); Sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Force = mg (1); Displacement = h (1); W = mgh (1); Stored as PE (1); Units (1).	1+1+1+1+1
	OR: P = 50 W (1); Energy = 6000 J (1); 1 kWh = 3.6 x 10^6 J (1); 6000/3.6 x 10^6 (1); = 0.00167 kWh (1).	1+1+1+1+1
20	PE 500 J (1); KE 500 J (1); v = sqrt(200) (1); = 14.1 m/s (1); Conservation (1).	1+1+1+1+1
	OR: Statement (1); W = change in KE (1); Example (1); Calculation (1); Units (1).	1+1+1+1+1

Mistakes to avoid:

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- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2 kWh	1 x 2.
2	**(c)** kinetic energy	Motion.
3	**(a)** load/effort	Formula.
4	**(c)** 4	L/E.
5	**(c)** potential energy	Elastic.
6	**(b)** first class lever	Fulcrum in the centre.
7	**(a)** potential to kinetic	Falling.
8	**(a)** 3.6 x 10^6 J	Commercial unit.
9	**(c)** created or destroyed	Conservation.
10	**(b)** maximum PE	Top.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	W = Fs (1); = 30 J (1).	1+1
12	Energy due to position or state (2).	2
13	Lever, pulley (1); Inclined plane (1).	1+1
14	mgh (1); = 120 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	mgh (1); = 300 (1); J (1).	1+1+1
16	kWh = 2 x 3 (1); = 6 (1); units (1).	1+1+1
17	Less force needed (1); Longer path (1); Same work (1).	1+1+1
	OR: P = W/t (1); 600/20 (1); 30 W (1).	1+1+1
18	MA = L/E (1); 5 (1); No unit (1).	1+1+1
	OR: Joule (1); Watt (1); kWh is a commercial unit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Class 1 (1); Class 2 (1); Class 3 (1); Examples (1); MA (1).	1+1+1+1+1
	OR: P = W/t (1); Unit watt (1); 1 W = 1 J/s (1); kW (1); Example (1).	1+1+1+1+1
20	Fixed (1); Movable (1); MA (1); Uses (1); Combination (1).	1+1+1+1+1
	OR: Direction (1); Force (1); Speed (1); Convenience (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

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